Integrating Computational Modeling and Patient Data to Reveal Regulatory Pathways in Alzheimer's Disease Neuroinflammation

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MEDH-205

Abstract

Background: Alzheimer's disease is a progressive neurodegenerative disorder with two prominent features: **chronic neuroinflammation** and the presence of **Aβ plaques** [1]. It is known that several **genetic risk factors,** such as **TREM2, IL1B, and ABCA1**, can affect the risk of developing AD; however, our understanding of the complex interactions within the broader neuroinflammatory system remains unclear [2,3].

Objective: This study addresses how **neuron-glia interactions** contribute to neuroinflammation in AD and how a **Boolean model** can identify key regulatory pathways [4]. We hypothesized that computational simulations will reflect known biological outcomes, such as **reduced immune activation after TREM2 knockout** or **increased Aβ buildup after ABCA1 loss**.

Methods: A qualitative **Boolean network model** of neuroinflammatory signaling was constructed in **GINsim (v3.0.0)** [5]. The model's logical rules were built using **gene expression data from published AD mouse knockout studies** (TREM2, IL1B, ABCA1, APOE) and literature review. This computational model was then combined with **human patient biomarker and gene expression data from the ADNI and AMP-AD databases** to link simulations to clinical disease progression. The Boolean neuroinflammatory network model predicts **sustained activation of microglial and pro-inflammatory nodes**, along with **reduced neuronal maintenance node activity**, reflecting the neuroinflammatory patterns observed in Alzheimer's disease [6]. We performed in **silico knockouts (KOs)** by setting target nodes to 0 and analyzing the resulting states, or "**attractors**," which represent long-term cellular conditions.

Results: Our results showed that simulated gene knockouts produced distinct outcomes. The **absence of ABCA1** has resulted in a **dramatic increase in glial-cell activation markers** (such as GFAP and TREM2), indicating that **ABCA1 generally inhibits the activation of glial cells**. The **absence of IL1B results in reduced levels of several inflammatory** markers, further supporting its role as a **principal mediator of inflammation**.

Results

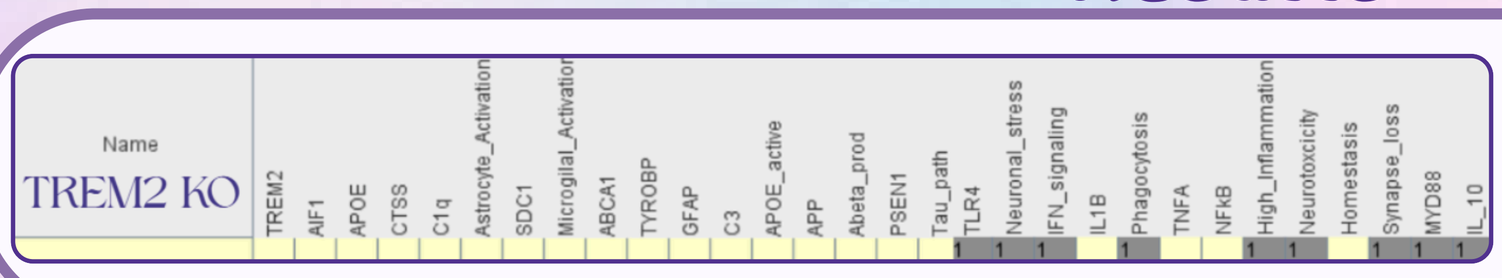


Figure 1. TREM2 KO) *TREM2 knockout significantly affects the status of several pathways and genes related to neuroinflammation and neurodegeneration, as marked by "1." This deletion primarily disrupts inflammatory signaling (TLR4, IL1B) and is associated with observable changes in phagocytosis, neurotoxicity, synapse loss, and inflammatory response (MYD88, IL10).*

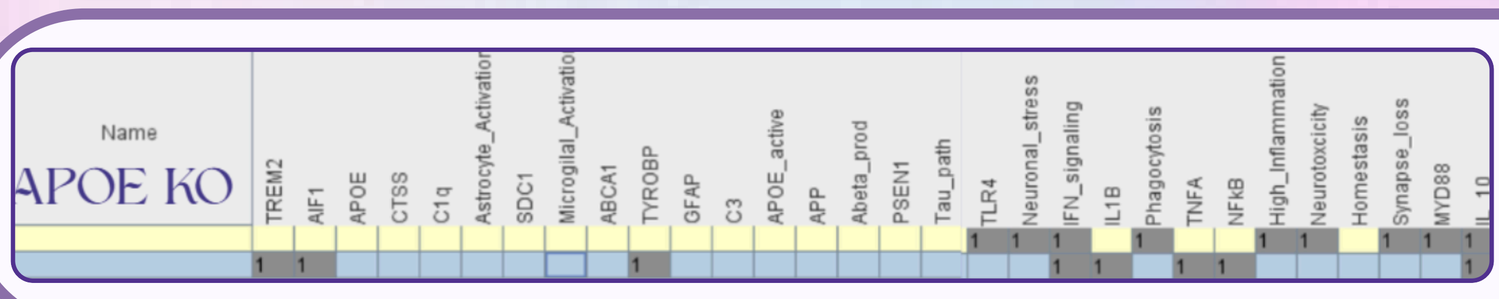


Figure 2. APOE KO) *APOE knockout resulted in significant alteration of numerous genes and pathways involved in neuroinflammation and neurodegeneration (marked by "1"). Specifically, the loss of APOE impacts glial cell activation, amyloid processing, inflammatory signaling, neurotoxicity, and synapse loss.*

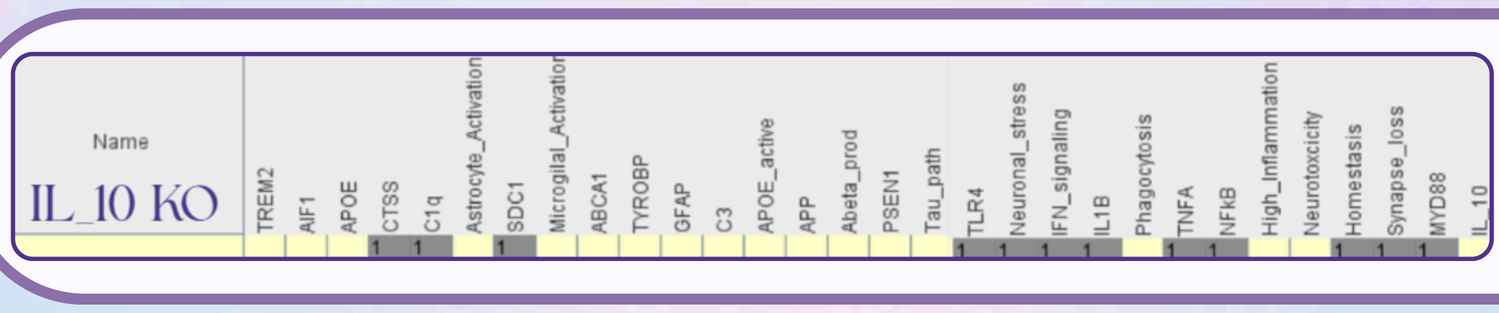


Figure 3. IL_10 KO) *The figure highlights the broad functional impacts of removing the anti-inflammatory gene IL_10 (marked "1"). This deletion leads to an increase in complement factors (C1q), microglial activation and is associated with multiple changes across inflammatory, neurotoxic, and synapse integrity pathways.*

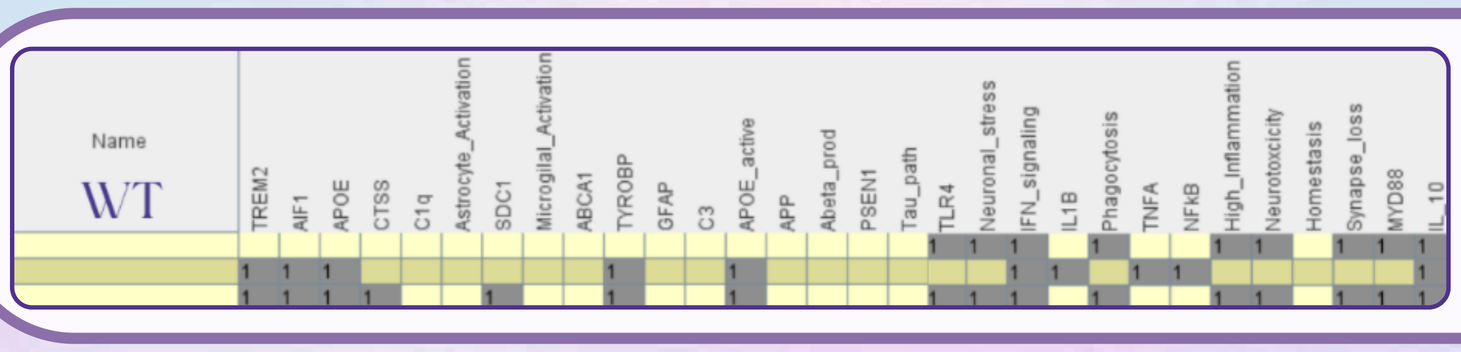


Figure 4. WILD TYPE) *This figure establishes the baseline activity for genes and pathways in a normal wild-type system, where "1" denotes active or measurable expression under control conditions. It serves as the reference state to assess the relative impact of the various gene knockout experiments presented in other figures.*

Knockout Model

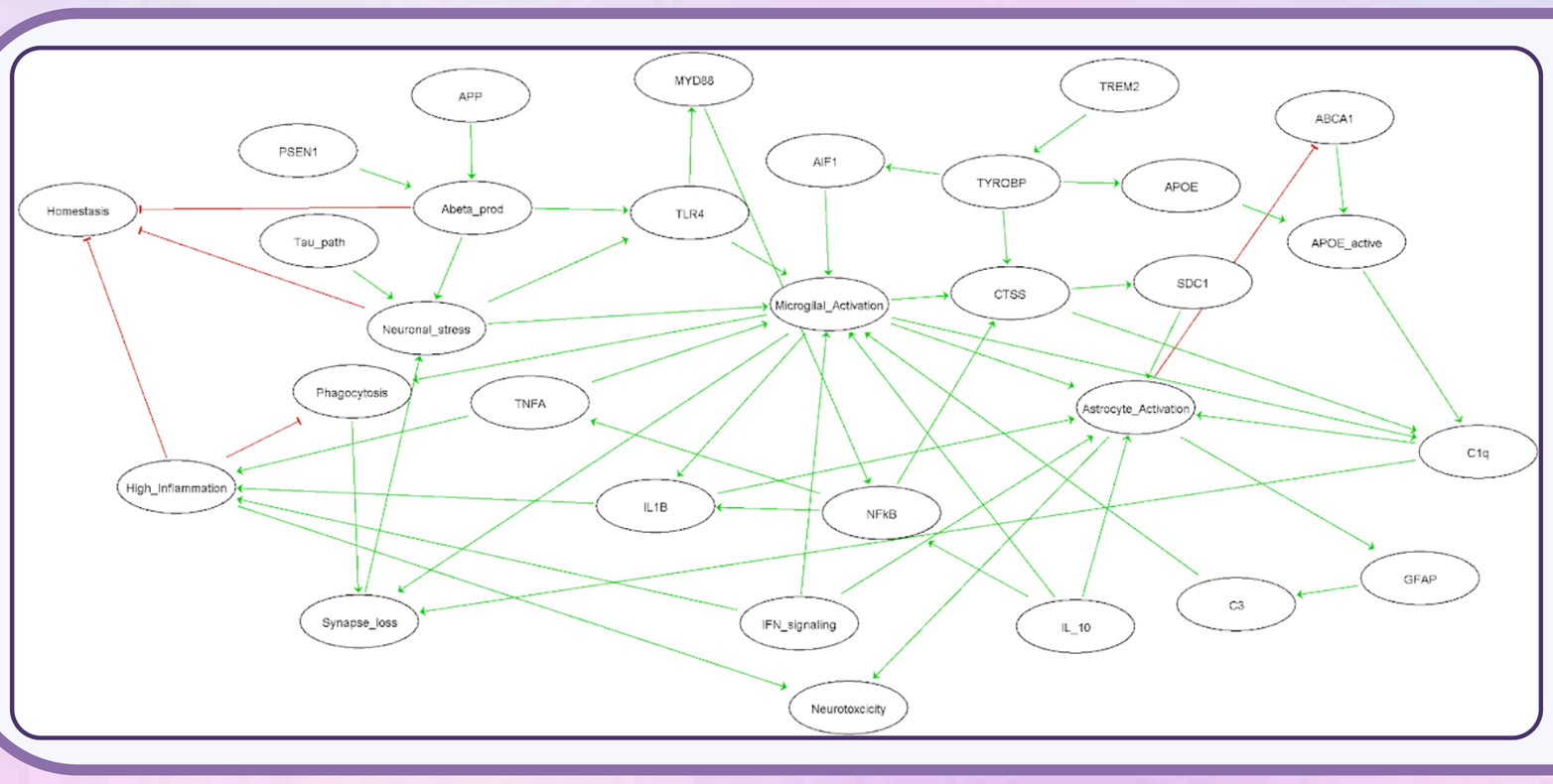


Figure 5. *This figure shows a logical regulatory network model constructed and simulated in GINsim, illustrating the effects of a gene knockout. Nodes represent key regulatory components, and edges indicate activating (→ in green) or inhibitory (⊣ in red) interactions. The knockout condition was implemented by fixing the targeted node to an inactive state (0), preventing its expression across all simulations.*

Research Question

How do **neuron-glia interactions**, particularly between *astrocytes and microglia*, contribute to **neuroinflammation** in **Alzheimer's disease**, and how can a Boolean model incorporating **gene expression data** and **patient data** help **identify key regulatory pathways**?

We expect that simulations using Boolean logic will **reflect known biological outcomes**, such as **reduced immune activation** following TREM2 deletion or **increased Aβ buildup** after ABCA1 loss.

Methods

1 Data Collection

We gathered **gene-expression datasets** from AD mouse knockout studies (*TREM2, IL1B, APOE, ABCA1*) and organized each study's gene **symbol**, **function**, **tissue type**, **age**, **differential expression**, and **accession numbers**.

2 Expression Profiling

Genes were classified as **up- or down-regulated**, and **fold-change** or **significance values** were noted when available. **Repeated patterns** across knockouts were used to identify **potential shared pathways** or **regulatory effects**.

3 Model Building

A qualitative **Boolean network** was constructed in **GINsim** using **knockout findings**, **pathway databases**, and **literature evidence**. *Nodes* represented **genes** or **processes**, and *logical rules* mapped **activating** or **inhibitory interactions**.

4 Patient Integration

Transcriptomic and **clinical data** from ADNI, ssREAD, and AMP-AD were **aligned with model nodes**. Each patient received **ON/OFF node** states based on gene expression or **machine-learning inference** to generate **personalized Boolean simulations**.

Subject ID	Diagnosis	Aβ42 (pg/mL)	Tau (pg/mL)	Hippocampal Vol (mm³)	MMSE	Amyloid Node	Tau Node	Predicted Outcome	Match
100_S_1000	CN	885.6	191.4	6944	30	OFF	OFF	neuroprotective	Yes
101_S_1010	CN	970.2	265.7	7493	30	OFF	OFF	neuroprotective	Yes
102_S_1020	CN	989.5	227.2	7345	30	OFF	OFF	neuroprotective	Yes
103_S_1030	CN	913.1	148.6	6616	30	OFF	OFF	neuroprotective	Yes
104_S_1040	CN	826.2	149.3	7123	28	OFF	OFF	neuroprotective	Yes
105_S_1050	CN	1031.1	201.6	6705	29	OFF	OFF	neuroprotective	Yes
106_S_1060	CN	868.8	224.5	7436	28	OFF	OFF	neuroprotective	Yes
107_S_1070	MCI	563.9	231.3	6322	26	ON	OFF	synapse loss	Yes
108_S_1080	MCI	383.6	190	5921	27	ON	OFF	synapse loss	Yes
109_S_1090	MCI	415.6	373.3	6197	28	ON	ON	pro-inflammatory	No
110_S_1100	MCI	706.9	237.5	6067	25	OFF	OFF	neuroprotective	No
111_S_1110	MCI	653.3	341.9	4563	25	OFF	ON	synapse loss	Yes

Patient Data Biomarker

Figure 6. Overview of subject-level clinical and biomarker data used for model evaluation**. Variables include diagnostic category, cerebrospinal fluid Aβ42 and tau concentrations, hippocampal volume, and Mini-Mental State Examination (MMSE) scores. **Binary indicators** reflect amyloid and tau node activation within the computational model. Predicted outcomes are compared against ground-truth labels in the "Match" column, allowing **assessment of model agreement across cognitively normal and mildly impaired participants.

5 KO Simulation

In-silico knockouts were created by **fixing nodes to 0** and **running asynchronous simulations across multiple seeds**. Knockout outcomes were scored using **Health and Inflammation metrics** and compared to **experimental KO effects**.

6 Statistical Testing

Agreement between **model predictions** and **experimental results** was assessed with **Fisher's exact test** and **FDR correction**. Additional tests (Kruskal-Wallis, Spearman, regression) **evaluated links between simulated states**, **biomarkers**, and **diagnostic groups**.

Results

• **APOE** and **ABCA1** knockouts consistently produced **strong activation** of **cytokine pathways** (TNFα, IL1B, NF-κB) and persistent **microglial** and **astrocytic activation**, indicating that **lipid-regulating genes** normally suppress **chronic neuroinflammation**.

• **TREM2** and **IL1B** deletions led to partial disruption of **inflammatory signaling**, with **TREM2 KO** shifting microglia toward a **pro-inflammatory**, low-phagocytosis state, while **IL1B KO** reduced specific cytokine outputs but did not eliminate downstream **neurotoxicity**.

• **IL-10 knockout** generated the most severe **inflammatory phenotype**, showing maximal activation of microglial markers, cytokines, and neurotoxic outputs, and **eliminating the protective attractor state**, highlighting **IL-10's essential anti-inflammatory role**.

• **TLR4 knockout** uniquely reduced multiple inflammatory components and **lowered neurotoxicity** and **synapse loss**, identifying **TLR4** as a major **upstream driver of inflammation**, consistent with activation patterns observed in human **AD transcriptomic datasets**.

Discussion

Our **Boolean network results** reflect known **neuroinflammatory mechanisms** in **Alzheimer's disease**. Strong inflammatory activation in the **APOE** and **ABCA1** knockouts aligns with studies linking these genes to impaired **lipid handling** and heightened **glial activation** [1]. Reduced **cytokine signaling** in the **IL-1β knockout** also matches its established **pro-inflammatory role.** Notably, **IL-10 deletion** produced the **most severe and persistent inflammatory state,** supporting the idea that **IL-10** is a **central regulator** that restrains chronic **microglial activation**. Overall, the model reproduces key **AD-related patterns** found in **patient transcriptomic datasets** and highlights **IL-10** as a potentially influential node in **neuroimmune control** [4].

Limitations

- Boolean **ON/OFF states** simplify complex signaling and may overlook graded effects (transitioning between binary states 0 or 1)
- **Mouse datasets** vary in strain and brain region, adding **inconsistency** to model inputs.
- **Human datasets** capture only part of **AD heterogeneity**.

Follow-up Studies

- Experimentally test **IL-10 knockout** effects in multiple AD **mouse models**.
- Compare **IL-10–related pathways** across strains to assess **consistency**.
- **Expand modeling** using multi-state or **quantitative frameworks**.
- Further examine IL-10's **downstream pathways** to clarify its regulatory role in neuroinflammation.

References

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Acknowledgements

This work was supported through the expertise provided by our mentors, Professor Shen and Dr. Chuang and guidance from our research suprervisor Dr. Davis